a0d22f89-8103-411a-9908-fd9be0e358a6

<prev_user>

<request>

You are working on an issue in the 'BlackbirdWorks/gopherstack' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/gopherstack/gopherstack (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-04-12T21:47:09.896Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section details on the original issue you should resolve

<issue_title>Shield: Implement Missing AWS SDK v2 Operations</issue_title>
<issue_description># shield Realism Gaps

The following AWS SDK v2 operations are missing from Gopherstack's `InMemoryBackend` and `shield` handler:

- [ ] Task 1: Implement AssociateDRTLogBucket
- [ ] Task 2: Implement AssociateDRTRole
- [ ] Task 3: Implement AssociateHealthCheck
- [ ] Task 4: Implement AssociateProactiveEngagementDetails
- [ ] Task 5: Implement CreateProtectionGroup
- [ ] Task 6: Implement DeleteProtectionGroup
- [ ] Task 7: Implement DeleteSubscription
- [ ] Task 8: Implement DescribeAttack
- [ ] Task 9: Implement DescribeAttackStatistics
- [ ] Task 10: Implement DescribeDRTAccess

Implementation Guide

To implement these missing operations natively:

1. **Interface Expansion (`services/shield/interfaces.go`)**
   Add the missing method signatures to the `StorageBackend` interface mapping directly to the AWS SDK client interfaces.

2. **Memory Backend Implementation (`services/shield/backend_memory.go`)**
   Create necessary struct mapping representations for these operations and track them in memory dictionaries. Ensure all mutations are thread-safe and protected under `b.mu.Lock()`.

3. **HTTP Routing (`services/shield/handler.go`)**
   Add the target mappings in the main JSON/XML routing switch block. Extract the payload accurately into the SDK `Input` struct, execute the backend method, and serialize the `Output` struct seamlessly. Include the supported strings in `GetSupportedOperations()`.

4. **Integration Testing (`services/shield/handler_test.go`)**
   Write authoritative table-driven integration tests verifying the exact AWS schemas, parameters, constraints, and edge cases.
</issue_description>

<agent_instructions>Please start working on issue #{issue_number}: {issue_title}.
{issue_url}

Please make a pull request with your changes. Please commit and push often so you don't lose work.</agent_instructions>

Comments on the Issue (you are @copilot in this section)

<comments>
<comment_new><author>@agbishop</author>
copilot-autodev: agent task created for issue BlackbirdWorks/gopherstack#925 (initial invoke).

</comment_new>
</comments>
</problem_statement>

Implement the necessary changes to the repository so that the requirements specified in the <problem_statement> are met.

<prev_assistant>

All 10 missing Shield operations have been successfully implemented. Here's a summary of what was done:

Summary

New Operations Implemented

1. **AssociateDRTLogBucket** - Associates S3 log buckets with the DDoS Response Team
2. **AssociateDRTRole** - Associates an IAM role with the DRT
3. **AssociateHealthCheck** - Associates a Route 53 health check with a protection
4. **AssociateProactiveEngagementDetails** - Stores emergency contact details for proactive engagement
5. **CreateProtectionGroup** - Creates a protection group with aggregation/pattern configuration
6. **DeleteProtectionGroup** - Deletes a protection group
7. **DeleteSubscription** - Cancels the active Shield Advanced subscription
8. **DescribeAttack** - Returns details of a specific attack event
9. **DescribeAttackStatistics** - Returns attack statistics summary
10. **DescribeDRTAccess** - Returns the current DRT access configuration

Changes Made

- **`interfaces.go`**: Added 10 new method signatures to `StorageBackend` interface
- **`backend.go`**: Added new types (`EmergencyContact`, `DRTAccess`, `ProtectionGroup`, `Attack`, `AttackStatistics`), updated `InMemoryBackend` struct with new maps/fields, implemented all 10 new backend operations
- **`persistence.go`**: Updated snapshot/restore to persist all new fields
- **`handler.go`**: Refactored dispatch into sub-dispatchers (fixes cyclop complexity), added HTTP routing for all 10 new operations, updated `GetSupportedOperations` (10→20 ops), changed nil-return handlers to return just `error` (fixes unparam lint issues)
- **`export_test.go`**: Added `ProtectionGroupCount` and `AttackCount` test helpers
- **`sdk_completeness_test.go`**: Removed 10 newly-implemented ops from `notImplemented` list

- **`handler_new_ops_test.go`**: Comprehensive table-driven tests for all 10 new operations

Quality Metrics
- **Coverage**: 92.7% (well above the 85% minimum)
- **Lint**: 0 issues
- **Race detector**: Clean

<current>

<request>

You are working on a task in the 'BlackbirdWorks/gopherstack' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/gopherstack/gopherstack (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-04-12T22:52:25.318Z</current_datetime>

Consider the following problem statement:

<problem_statement>
(refinement check 1 of 3 against issue #925). Please review your implementation against all requirements in the original issue and refine anything that is missing or incomplete. While refining look for any missing features compared to localstack and aws realism. Find any performance issue or optimizations along with any issues with concurrency. Ensure there is no resource leaks or go routine leaks. Find at least 20 additional items in each refinement, commit and push often. Please address any review feedback.
</problem_statement>

First, determine whether the <problem_statement> requires code changes, an explanation, a plan, or analysis. Then follow the appropriate path:

If the task requires code changes:

**If the task is a question or only requires creating a plan or

```
   analysis:** respond directly without making code changes.
```